

The Evolutionary Journey of NLP in Aspect-Based Sentiment Analysis: Evaluating Traditional Techniques, Deep Learning, and LLMs

CS21

Final Report



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CONTRIBUTION STATEMENT

Our group CS21, taking project titled The Evolutionary Journey of NLP in Aspect-Based Sentiment Analysis: Evaluating Traditional Techniques, Deep Learning, and LLMs, with group members Albert Huang, HanChen Zou, Weiguang Lu, Austin Mi, Jiajian Pang, Prem Kumar Rajendran, would like to state the contributions that each group member has made for this project during this semester:

- group member 1 Albert Huang: Project manager, managing the entire project, making group-client-tutor-teaching team communication, conducting work distribution, planning, progress tracking, risk management and all other project management related activities. Researcher, conducted researches on XAI, ABSA that the project needed. Recorder & reporter, managing all documentations related to this project and write reports on meeting, group-level documents like checking forms, large parts of proposal, scientific report & final report. Data analyst, collected project results and conducted key analysis.
- group member 2 HanChen Zou: Designed and implemented a multi-task model based on T5 + LoRA for ABSA. Developed a two-head classifier for biomarker and sentiment classification base on T5. Integrated XAI modules for T5 (LIME, saliency and attention visualization). Completed T5 performance evaluation, visualization report and technical documentation writing.
- group member 3 Weiguang Lu: Programmer, responsible for the development and practice of Transformer-based models (BERT and RoBERTa) in the project. Researcher, responsible for finding relevant literature and evaluating how to reasonably apply XAI to the model. Data analyst, responsible for training the model and obtaining results, analyzing and processing the results, and optimizing the model based on the results; data processing, localizing the data to avoid the impact of the imbalance of the data set on the classification results. Recorder, editing and writing reports.
- group member 4 Tianxing Mi: Mainly responsible for the code deployment of deep models based on RNN, LSTM, GRU and BiLSTM, and optimized for specific tasks. In-depth review of relevant literature, analysis of existing models and application of

XAI technology to enhance the transparency of the model. Responsible for training these four models and visualizing the results. Also optimized the hyperparameters of the data to improve the generalization ability of the model. Wrote the project report of the deep learning part, recorded the research process and results, and ensured the accurate communication of information.

- group member 5 Jiajian Pang: Throughout the project, I was primarily responsible for designing, developing, and evaluating the Traditional NLP component. I implemented both SVM and Decision Tree classifiers on the SemEval 2014 and 2015 datasets, including extensive parameter tuning and cross-dataset performance comparison. I also integrated explainability tools such as LIME and SHAP to enhance model interpretability. In addition to development, I wrote the entire Traditional NLP section of the final scientific report, including model justification, mathematical formulation, and evaluation visualizations. I participated in client meetings, contributed to demo video production, and actively supported team coordination for final reporting and presentation. My work ensured that the traditional NLP pipeline was not only functional but also transparent, well-documented, and analytically rigorous.
- group member 6 Prem Rajendran: Data Exploration and selected datasets based on the project requirements. Designed, analyzed and implemented data pre-processing for the models. Source code integration for all modules, CICD (GITHUB repository management), integration testing and report visualization and results validation and comparison. Prepared and summarized project documentation, weekly team presentation, and presented with tutor and client to showcase the project progress and team's contributions and gathered their valuable feedback and collaborated with the team for managing and implementing their recommendation changes.

All group members agreed on the contributions listed on this statement by each group member.

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Abstract

Aspect-based Sentiment Analysis (ABSA), a specific subfield of Sentiment Analysis (SA) within the wide border of Natural Language Processing (NLP), focusing on tasks like aspect-extraction and opinion mining, is an area that has been evolving for years. With the rising need from the application of customer feedback gathering, public opinion poll conducting and more, Aspect-based Sentiment Analysis has gained increasing attention for its capability and potential. It is observed that more and more new researchers have entered the field, a quick guide of ABSA emphasized on technique evolution would help them gain proper understanding of the field, lowering entrance barricades, and overall benefit the growth of the field. Thus, this project is built to providing such a guide on the evolution of Aspect-based Sentiment Analysis. The project contains a literal research on the topic and a comparative analysis on the performance of several NLP models mainly in 4 categories: traditional NLP, deep learning, transformer-based and Large Language Models using two selected datasets. And these two parts are eventually connected to form a detailed scientific report as final product. Explainable AI (XAI) and customized visualizations are adopted in all models to provide experimental interpretability improvement along with model structure graphs and tables of various usages showing in the scientific report. The project finds out that advanced models like transformer-based and LLMs have the overall better performance, and the timeline of ABSA evolution is built.

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Introduction

Sentiment-based Aspect Analysis (ABSA) performed on text data which circles around a common topic, an entity. And from the content within those text data, it extracted involved aspects of that entity. Then, it provides sentiment polarity attached to those aspects. Tasks, terms and classification of ABSA has been evolving over the years through different studies, from two subtasks ideologies: aspect extraction and aspect sentiment classification (Liu 2012), to three subtasks ideologies: aspect term extraction, aspect term aggregation, and aspect term polarity estimation (Pavlopoulos 2014), or from the perspective of number of elements, single ABSA tasks and compound ABSA tasks (Zhang et al. 2022), the branches of ABSA grow. Selecting the right section of ABSA techniques and architecture would be vital for any real-world practice involving ABSA.

As the field prospering grows, and studies overlap, the entrance barrier gets higher, blocking new researchers into the field or slower the progress of absorbing new resources into the field. A comprehensive review of the field is needed, and there have been various attempts. Those studies and more (explain in next chapter related literature), together built a solid foundation for ABSA systematic review and techniques analysis, and gave inspiration to this project.

Different from those related projects and studies, this project would: One, provide a entrance-level-understandable interoperability guide on the topic. Two, combine practice analysis of modern models used in ABSA and literal academic review. Three, experimentally use Explainable AI in all models to explore on model aspects.

The project would combine those three points as guide, to make an attempt to give an systematic review of ABSA evolution on NLP techniques and a comparative analysis on iconic ABSA NLP models' performance using two selected dataset. And in the end, produce a scientific report that concludes the entire project as its final product. Key sections of the scientific report are merged into this final report for a better demonstration on the project.

Related Literature

2.1 Literature Review

2.1.1 Introduction

The project demands ongoing researches on the topics as the project develops to help provide solid background studies and academic supports. In this section, a concise version of the literature review is shown. This reviews application fields of ABSA, similar and related studies like this project, and the evolution of ABSA's NLP techniques. We prioritized research papers that are well-known, most-cited or from authoritative academic institution as prime source. Only studies with related topics are considered.

2.1.2 Review on related studies & projects

The growing techniques of ABSA supported the increasing need, more sophisticated task demands and specifications of ABSA application. For instance, opinion gathering services like political public opinion polls now require specific emotion detection and analysis instead of simple sentiment polarities analysis and this can be seen in many studies of political opinions (Tanoto et al. 2024; Gold et al. 2018; Ahmad et al. 2017). In e-commerce, analyzing product review would require compound ABSA tasks, like having multi-element pairs (Dogra et al. 2024; Princess et al. 2024). It is common for a user's review, like "The screen is awesome, but the battery is not good." to have multiple aspects and sentiments within one single statement. Or considering the background of globalization, multi-language support (Maghsoudi et al. 2025) is a crucial need for any international company's opinion gathering system. Multilingual and cross-cultural supported ABSA would be hard to achieve with limited dataset using traditional NLP models, however, addressed easily with the adoption of Large Language Models like Llama 3 or generative Artificial Intelligence (AI) like GPT

(Mathebula et al. 2024; Pan et al. 2023). Conversely, the enlarging application field in various industries stimulate the further exploration and development of the ABSA techniques. Only three years after the emergence of the booming AI trends and its AI application, thousands of research papers aiming the combination of ABSA with the latest AI technologies had been released. The techniques of ABSA had been pushed to an all-new edge in the background of the AI market.

In 2020, in the study of “Issues and Challenges of Aspect-based Sentiment Analysis: A Comprehensive Survey” (Nazir et al. 2020) by Nazir and others, ABSA is reviewed based on the issues and challenges faced in the form of questions (Nazir et al. 2020). Three main issues are transformed into three main tasks: Aspect Extraction, Aspect Sentiment Analysis and Sentiment Evolution followed by subtasks within these sections (Nazir et al. 2020) . In 2022, another study “A Survey on Aspect-Based Sentiment Analysis: Tasks, Methods, and Challenges” (Zhang et al. 2022) by Wenxuan Zhang and others, followed the study of Liu, and expand the idea of sentiment element into four: aspect category, aspect term, opinion term and sentiment polarity (Zhang et al. 2022). Based on these four elements, the study defines ABSA as a problem of identifying these elements. The study summarizes and categorizes all tasks (up to 2022) into two kinds: single ABSA tasks involving only one element, and compound ABSA tasks with multiple elements (Zhang et al. 2022). In this study of Zhang, it provides a new aspect of defining ABSA and its tasks with a structured terminology. These two studies present and review ABSA on the structure level of tasks. In the study of “Exploring Aspect-Based Sentiment Analysis Methodologies for Literary-Historical Research Purposes” (Dejaeghere et al. 2024) by Dejaeghere and others in 2024, literature review in a timely order is conducted and ABSA pipeline development with comparative analysis are discussed (Dejaeghere et al. 2024). However, no practice experiment is involved in this study. And there is a study like “Survey on ABSA based on machine learning, deep learning and transfer learning approach” (Kalbhor and Goyal 2023) , which is conducted in a similar selection of aspects like this project.

Traditional NLP were widely used in ABSA before the rise of deep learning. In the era of traditional NLP, the tasks of ABSA were relatively simple using limited data volume under non-complex analysis scenarios like (Menzli 2023). Various traditional NLP techniques were used at the time, iconic examples are : rule-based sentiment dictionaries or statistical methods like N-Gram models. Later on, classic machine learning methods like Naïve Bayes

classifiers (DataCamp 2023; De La Cruz 2023) are used in ABSA tasks. These methods generally contain manually constructed features like word frequency, part of speech, sentiment word matching. . . The outcome would be a simple polarity prediction on the related aspect. Though using these methods, a desirable outcome can be guaranteed if carefully constructed the involved feature, there are many drawbacks and limitations. These models have a very limited flexibility in terms of output format, cross-domain adaptability, and text environment (Apify 2023; Kumar and Sebastian 2012) They also lack of ability for clearly differentiating ambiguous polysemous words, processing complex texts (Apify 2023).

Early deep learning approaches on ABSA can be traced back to 2015 and 2016, they mainly explored on identifying aspects and polarity of sentiment with studies in features like linguistic features, grammatical relations, machine learning classifiers and topic modelling (Schouten and Frasincar 2015). And later on, with the development of deep learning and the studies of Deep learning in ABSA occurred, there are systematic reviews on this topic, which explore various deep learning models (Chauhan et al. 2023; Do et al. 2019). . Models like RecNN, DNN, LSTM are used and compared in terms of performance on certain datasets. Under our own research, we believe that the basic RNN model is not a suitable choice for ABSA of this study due to the characteristic of the prone of gradient disappearance and its short-term memory favor. LSTM and BiLSTM could be a more suitable choice since its advantages in long-term dependency tasks as ABSA requires medium- and long-term analysis (Siami-Namini et al. 2019; Staudemeyer and Morris 2019). BiLSTM should have better performance than LSTM since BiLSTM is able to capture bidirectional dependency, giving more context information in a more complex text environment (Siami-Namini et al. 2019).

Transformer-based approaches are new models that occurred in recent years. The self-attention mechanism allows the model to capture global context information and address long-distance dependency issues (Vaswani et al. 2017). In ABSA, these pre-trained models, like BERT, RoBERTa, can achieve high levels of performance through its unique mechanism (Jawahar et al. 2019; Liu et al. 2019), and can fine-grained model complex sentiment scenarios through multi-level feature fusion (Sun et al. 2019; Devlin et al. 2019).

Large Language Models and generative AI are advanced approaches that are being experimented and evolving at the moment. With the rise of AI significantly marked by GPT 3.5 to the world, many studies are being conducting, using various types of LLMs and generative AI models like GPT-3.5-turbo13, PaLM bison-001. . . (Mughal et al. 2024) and even some

hybrid approaches that contain the use of LLMs (Zhang et al. 2024; Negi et al. 2024). Each model has different focus, for example, T5 has a standard encoder-decoder structure, which can unifies multi-task learning with text generation (Raffel et al. 2020). As the rapid growth of this area, no definitive LLMs approaches has been recognized for ABSA tasks, and it is expected to be continuously changing and evolving for years to come.

2.1.3 Conclusion

The application field of ABSA is growing as its techniques evolved, and many studies have reviewed on ABSA using different perspectives and objectives. The evolution of NLP techniques in ABSA can be categorized into four sections, traditional NLP, deep learning, transformer-based and LLMs, each with its own developments.

PROJECT PROBLEMS

3.1 Project Aims & Objectives

This project aims to systematically review on the evolution of Aspect-Based Sentiment Analysis (ABSA) in terms of its NLP techniques, and conducted a comparative analysis on the performance of models using these NLP techniques in ABSA tasks on two selected datasets. And, explore on these models along the way. Models are selected from four main approaches: traditional NLP, deep learning, transformer-based and large language models with all approaches covered. The project produced a scientific report that concluded the project by connecting the review, practical experiments and comparative analysis.

3.2 Project Questions

As the project covers multiple models and approaches, the key questions we built the project around are: Is there a ultimate solution of models for all ABSA tasks? And if there is, what is it? If there is none, based on the result of this project, what models should be considered in what scenarios?

Besides that, there are also many other questions that help us design the project processes and develop project details. For example, how to ensure scientific report's readability to someone with no related knowledge? Or, how to manage a proper degree of performance-trade-off in all models when optimizing all models?

Countless project questions occurred during the development of this project. At the end of the project, some of the questions are addressed while some remains to be clear by future studies.

3.3 Project Scope

Constrained by reality conditions, this project has a limited scope with the following items:

- **Documentation:** A OneDrive folder that shared with our client containing all the related documents, separate code files, instructions files, project guide, a link to GitHub repository. The folder is well organized in a clear, professional format.
- **Data:** SemEval-2014 and SemEval-2015 are the two dataset used in this project. (Refer to section 4.2 and 4.3 for more details.
- **Implement Models:** **SVM & Decision Tree** for traditional NLP, **RNN, GRU, LSTM and BiLSTM** for deep learning, **BERT & RoBERTa** for transformer-based, and **T5** for LLM. All models are implemented with XAI experimentally to improve interpretability.
- **Scientific Report:** This is the product that addressing the project aims and objectives. In more details, the scientific report would first lead the audience through a comprehensive review on ABSA evolution by summarizing different sections of NLP techniques, and give a comparative analysis for these approaches along with a visually clear timeline graph. Then the scientific report would move on the practice side with methodologies being explained meticulously, from dataset to models, evaluation metrics and XAI implementation. The result of the practice will be carefully recorded and studied to provide a comparison analysis which would be expected to be a reasonable reflection on model performance when using a larger, border range of dataset.
- **Development Period:** The Project started from week 2 during semester 1 2015 and ended in week 13 semester 2025.

METHODOLOGIES

4.1 Methods

Overall, the project follows the flow of process below as the method on project development and conduct the scientific report. Ongoing Research & Literature Review are conducted along the project development to suit our changing needs. After we gain project requirements and needs from our client, we began Data Collection & Preprocessing on the two selected datasets. Then, we implemented models using these datasets. We gained results using a unified evaluation metrics and techniques, visualized them. At last, we combined our research and practical experiments to form a scientific report as final product delivered to client.

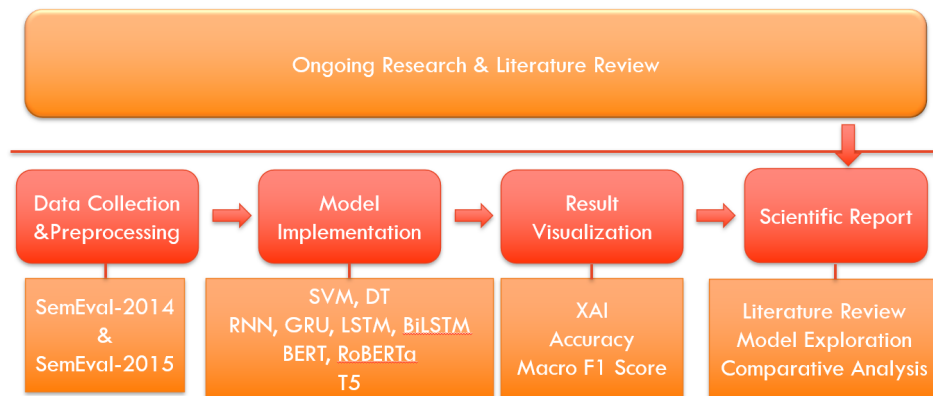


FIGURE 4.1: Project Process Flow

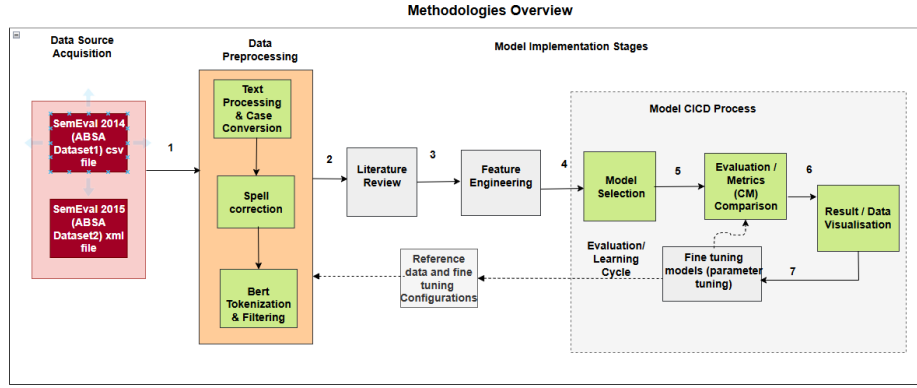


FIGURE 4.2: Methodology Technical Overview

In the following subsections, the details of each implemented models for the four approaches: traditional NLP, deep learning, transformer-based and LLMs will be explained with the explorations on each models.

4.1.1 Traditional NLP

In the traditional NLP component of this project, we selected Support Vector Machines (SVM) and Decision Tree (DT) as the core classification models. The decision was grounded in the structural suitability of these algorithms for sentiment classification tasks, particularly in handling the challenges commonly found in textual data: high dimensionality, sparsity, noisy labels, and nonlinear decision boundaries. Sentiment polarity classification is a typical text classification problem where input features (e.g., TF-IDF vectors) are sparse and often high-dimensional. Moreover, sentiment boundaries are not always linearly separable due to language ambiguity and context dependence. From a structural perspective, SVM and Decision Tree each address these challenges using distinct mechanisms.

- Support Vector Machine (SVM):** SVM aims to find an optimal hyperplane that maximizes the margin between different classes in the feature space. This formulation is particularly advantageous when working with high-dimensional and sparse data, which is typical of TF-IDF feature representations. Linear SVMs are computationally efficient and effective when the sentiment boundary is nearly linear. The inclusion of kernel functions also allows SVMs to handle non-linear separations by projecting data into higher-dimensional spaces. These properties make SVMs

a popular and proven choice for sentiment analysis, especially in early work using bag-of-words and TF-IDF features (Pang et al. 2002; Wahba et al. 2022).

- **Decision Tree (DT):** Decision Tree is a fundamental and intuitive supervised learning algorithm that splits data recursively based on feature values to make predictions through a tree-like model. Each internal node represents a decision based on a feature, and each leaf node corresponds to a predicted output. Its inherent interpretability makes it well-suited for applications where model transparency is essential. Although individual decision trees are prone to overfitting, their low computational cost and fast training make them valuable for tasks like text classification and sentiment analysis. Moreover, recent advancements in pruning techniques have improved their generalization performance (Quinlan, 1986; Rokach & Maimon, 2005).

Support Vector Machine

Support Vector Machine (SVM) is a binary classification model whose core objective is to find an optimal hyperplane in the feature space that separates instances of different classes with the maximum possible margin.

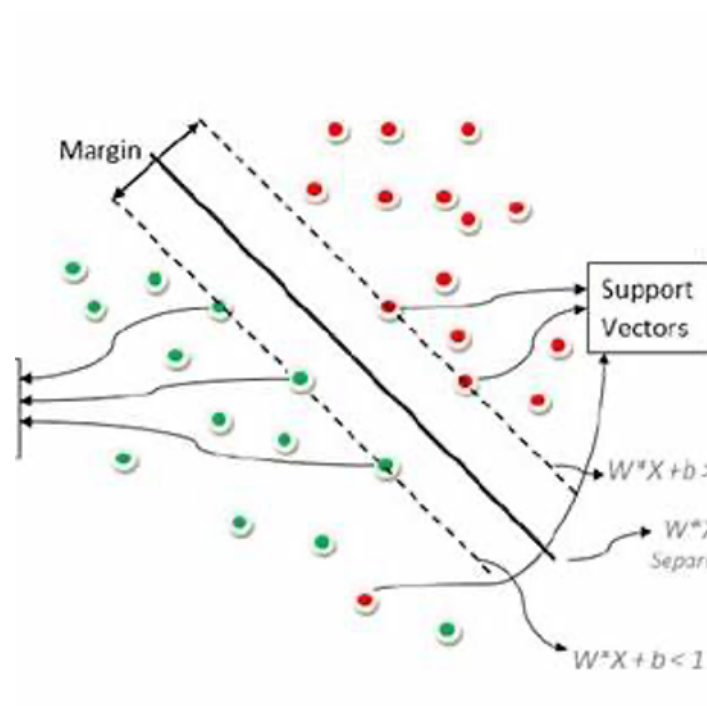


FIGURE 4.3: Illustration of the Optimal Hyperplane in a Linear SVM

A larger margin generally corresponds to better generalization. The basic mathematical formulation of a linear SVM can be expressed as the following constrained optimization problem:

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s.t.} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad \forall i \end{aligned}$$

Here, $\mathbf{w}$ is the normal vector to the hyperplane and determines its orientation, while b is the bias term that shifts the hyperplane. Each input feature vector is denoted as $\mathbf{x}_i \in \mathbb{R}^d$, and its corresponding label $y_i \in \{-1, 1\}$ represents the sentiment polarity class.

However, real-world sentiment analysis tasks often involve noisy and non-linearly separable data. To handle such cases, SVM introduces a soft-margin approach by incorporating slack variables $\xi_i \geq 0$, which allow certain data points to violate the margin constraints. A penalty parameter C is introduced to control the trade-off between maximizing the margin and minimizing the classification error. This leads to the following optimization problem:

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_i \xi_i \\ \text{s.t.} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \end{aligned}$$

In this formulation, ξ_i represents the extent to which the i -th training sample violates the margin, and C is a regularization constant that penalizes these violations. A large C places more emphasis on minimizing training error, which can increase the risk of overfitting. On the contrary, a smaller C allows for more violations, encouraging a wider margin and potentially better generalization.

In our project, we adopted a linear kernel SVM, which is well suited for sparse, high-dimensional TF-IDF feature representations derived from textual data. The model demonstrated strong performance in short-text sentiment classification tasks, offering efficient training and robust prediction accuracy.

Decision Tree

Decision Tree is a supervised learning algorithm that recursively partitions the input space based on feature values to construct a tree structure for classification or regression (Quinlan, 1986). Each internal node represents a decision rule on a feature, each branch corresponds to an outcome of the rule, and each leaf node represents a final prediction label or value. The algorithm's goal is to build a tree that best splits the dataset into homogeneous subsets with respect to the target variable.

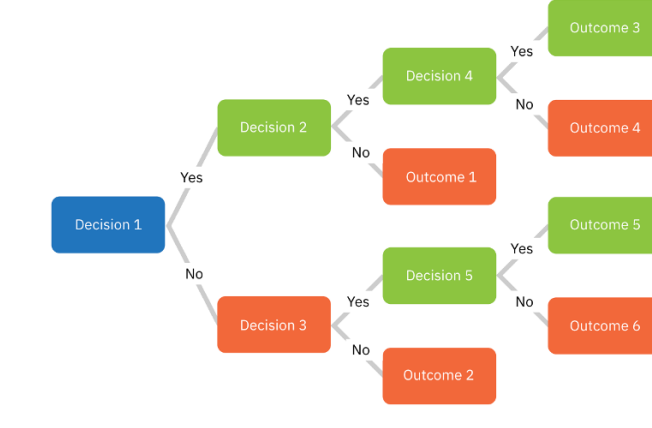


FIGURE 4.4: Illustration of the Decision Tree

The Gini Impurity, a common metric for split quality, is defined as: During training, the algorithm searches for the optimal feature and threshold to split the data at each node. This process relies on a splitting criterion that quantifies the impurity or disorder within a dataset. A commonly used impurity metric in classification tasks is the Gini Impurity, defined for a dataset D with C classes as (Breiman et al., 1984):

$$Gini(D) = 1 - \sum_{i=1}^C p_i^2 \quad (4.1)$$

where p_i is the proportion of instances in class i within the node. A lower Gini value indicates a purer node. At each candidate split, the algorithm evaluates the weighted average impurity of the left and right subsets and chooses the split that maximizes the reduction in impurity (Nowozin, 2012). The Gini Gain (also referred to as Information Gain when using entropy) is computed as:

$$Gain(D, t) = Gini(D) - \left(\frac{|D_{left}|}{|D|} \times Gini(D_{left}) + \frac{|D_{right}|}{|D|} \times Gini(D_{right}) \right) \quad (4.2)$$

This greedy, recursive partitioning continues until a stopping condition is met such as reaching a maximum depth, minimum number of samples at a node, or when further splits no longer improve purity.

Alternatively, Entropy can be used as the impurity measure, inspired by information theory (Wang et al., 2015). For a node D , the entropy is defined as:

$$Entropy(D) = - \sum_{i=1}^C p_i \times \log^2(p_i) \quad (4.3)$$

and the corresponding Information Gain for a split is:

$$Gain(D, t) = Entropy(D) - \left(\frac{|D_{left}|}{|D|} \times Entropy(D_{left}) + \frac{|D_{right}|}{|D|} \times Entropy(D_{right}) \right) \quad (4.4)$$

These splitting criteria ensure that each decision node moves the tree closer to subsets of high class purity (GeeksforGeeks, 2025). Decision trees are sensitive to overfitting, particularly when grown to full depth without pruning. Pruning techniques, such as cost-complexity pruning, are applied to reduce overfitting by removing subtrees that do not significantly improve generalization.

Due to their ability to model complex, non-linear feature interactions and offer human-readable decision rules, decision trees have proven effective in sentiment analysis tasks—especially for early-stage feature selection or as interpretable baselines in natural language processing pipelines.

4.1.2 Deep Learning

In the sequence modeling components of this project, we selected recurrent neural networks (RNN), long short-term memory networks (LSTM), bidirectional long short-term memory networks (BiLSTM) and gated recurrent units (GRU) as core time series classification models. They can effectively capture the temporal dependencies and contextual information between words in the text, and are particularly suitable for modeling long-range dependencies and nonlinear boundaries in sentiment polarity classification tasks.

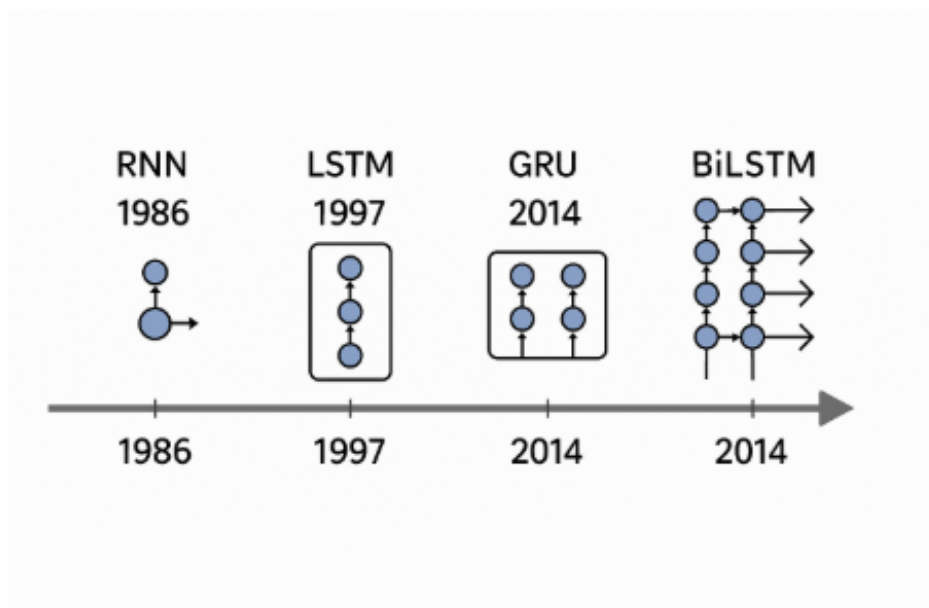


FIGURE 4.5: Deep Learning Model time line.png

From the timeline diagram, we can see that since the earliest recurrent neural network (RNN) came out in 1986, sequence modeling has taken a key step: RNN captures the near-term dependencies in the text by feeding back the hidden state of the previous moment to the current moment, but because the gradient decays or explodes in long sequences, it is difficult to learn long-distance information. (Elman 1990) In 1997, Hochreiter and Schmidhuber proposed the long short-term memory network (LSTM), adding "memory units" and multiple gates on the basis of RNN to dynamically determine the retention and forgetting of information, thereby significantly alleviating the gradient disappearance problem and allowing the model to robustly learn dependencies dozens or even hundreds of steps away. (Hochreiter and Schmidhuber 1997) In 2014, Cho et al. introduced the gated recurrent unit (GRU), which has an overall structure similar to RNN but similar logic to LSTM. It simplifies the three-gate structure of LSTM into an update gate and a reset gate, with fewer parameters, more efficient calculations, fast training speed and performance comparable to LSTM. (Cho et al. 2014) In the same year, bidirectional LSTM (BiLSTM) also became popular: it runs LSTM in two directions in parallel, concatenating the forward and reverse hidden states, which can not only utilize past information but also combine future context. Therefore, it performs particularly well in tasks that require complete sentence understanding, such as sentiment analysis and named entity recognition. (Schuster and Paliwal 1997)

The following is a detailed introduction to the structure of these models. The figure shows the basic structure of the four models.

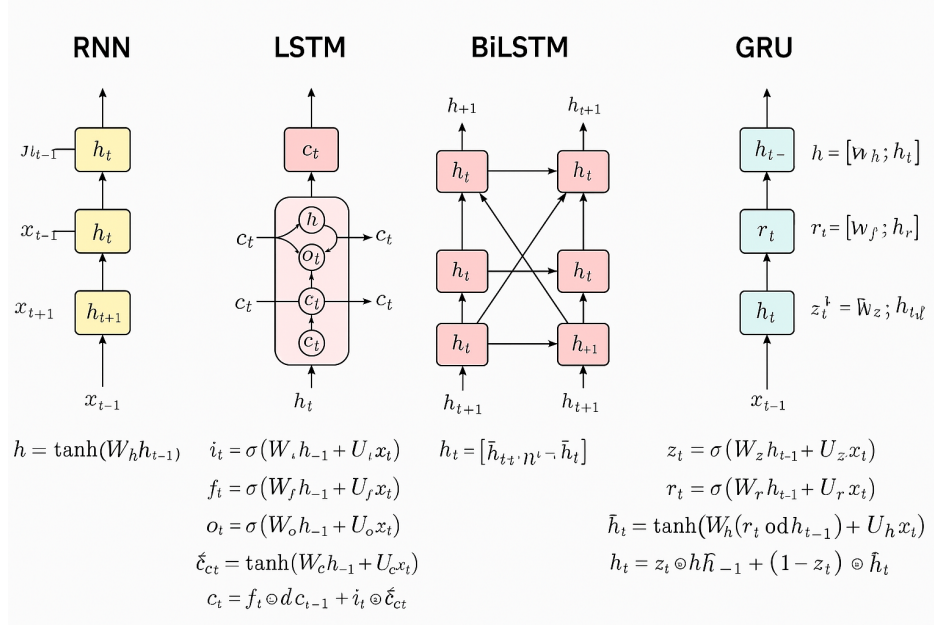


FIGURE 4.6: Model structure diagram

Recurrent Neural Network (RNN):

RNN is the most basic sequence model, and its core lies in the linear transformation of the previous hidden state h_{t-1} and the current input x_t , followed by nonlinear activation:

$$h_t = \tanh(W_h h_{t-1} + U x_t)$$

In this way, historical information can be passed down step by step. However, due to the gradient multiplication effect, RNN often encounters the problem of gradient vanishing or exploding when processing long sequences, resulting in insufficient capture of long-range dependencies. (Elman 1990)

Long Short-Term Memory (LSTM)

In order to solve the long-term dependency problem of RNN, LSTM introduces three gates at each moment (input gate i_t , forget gate f_t , output gate o_t) and a dedicated "cell state" C_t :

$$\begin{aligned}
i_t &= \sigma(W_i h_{t-1} + U_i x_t), \\
f_t &= \sigma(W_f h_{t-1} + U_f x_t), \\
o_t &= \sigma(W_o h_{t-1} + U_o x_t), \\
\tilde{C}_t &= \tanh(W_c h_{t-1} + U_c x_t), \\
C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \\
h_t &= o_t \odot \tanh(C_t).
\end{aligned}$$

In this way, LSTM can selectively "forget" or "retain" past information, significantly alleviating the gradient vanishing problem and performing well in long text sentiment analysis. (Hochreiter and Schmidhuber 1997)

Bidirectional LSTM (BiLSTM)

BiLSTM runs an LSTM in both the forward and reverse directions of time, concatenating the forward hidden state $\vec{h}_t$ with the backward hidden state $\overleftarrow{h}_t$ to obtain the final output:

$$h_t = [\vec{h}_t ; \overleftarrow{h}_t].$$

This structure can simultaneously utilize the contextual information before and after, which is particularly beneficial for sentiment classification that requires a full understanding of the overall semantics of the sentence. (Schuster and Paliwal 1997)

Gated Recurrent Unit (GRU)

GRU is a simplification of LSTM, retaining only two gates: Update gate z_t and reset gate r_t :

$$\begin{aligned}
z_t &= \sigma(W_z h_{t-1} + U_z x_t), \\
r_t &= \sigma(W_r h_{t-1} + U_r x_t), \\
\tilde{h}_t &= \tanh(W_h (r_t \odot h_{t-1}) + U_h x_t), \\
h_t &= z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t.
\end{aligned}$$

By merging the forget gate and the input gate into one update gate, GRU has fewer parameters and is more computationally efficient, achieving performance close to or better than LSTM in many NLP tasks. (Cho et al. 2014)

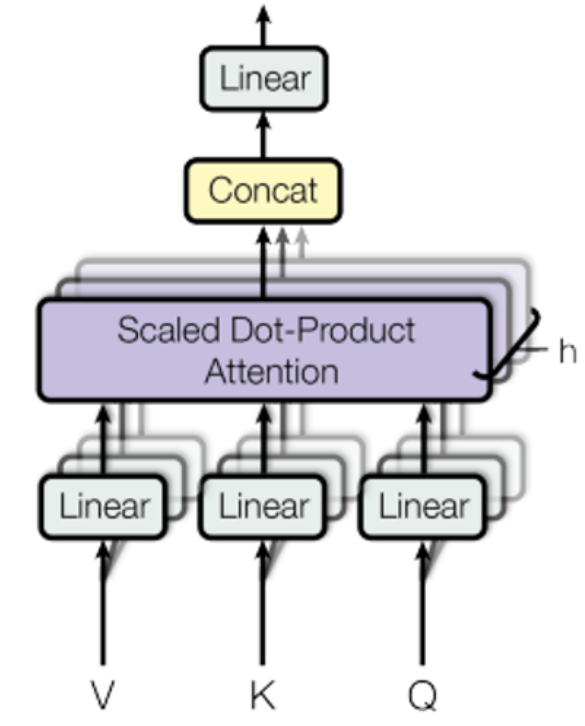


FIGURE 4.7: Multi-head attention (Vaswani et al., 2017)

4.1.3 Transformer-based

This section introduces the model construction and training methods for aspect-based sentiment analysis (ABSA) using BERT (Bidirectional Encoder Representations from Transformers) and RoBERTa (Robustly optimized BERT approach).

4.1.3.1 Transformer Architecture

The Transformer architecture is different from the previous RNN/convolutional network. It is better suited for sequence modeling. Its core ideas include: 1. Self-Attention: For each token in the sequence, score it according to its similarity with all other tokens, thereby dynamically aggregating contextual information; 2. Multi-Head Attention: Using multiple attention heads in parallel allows the model to learn diverse representations from different subspaces Multi-Head Attention: Using multiple attention heads in parallel allows the model to learn diverse representations from different subspaces (4.7); 3. Positional Encoding: Since there is no sequential recursion, sine/cosine functions are used to inject explicit position information for each position (function (4.5) and (4.6)); 4. Encoder-Decoder Structure: The encoder of stacked layers is responsible for understanding the input sequence, and the decoder generates the target sequence (4.8) (Vaswani et al. 2017).

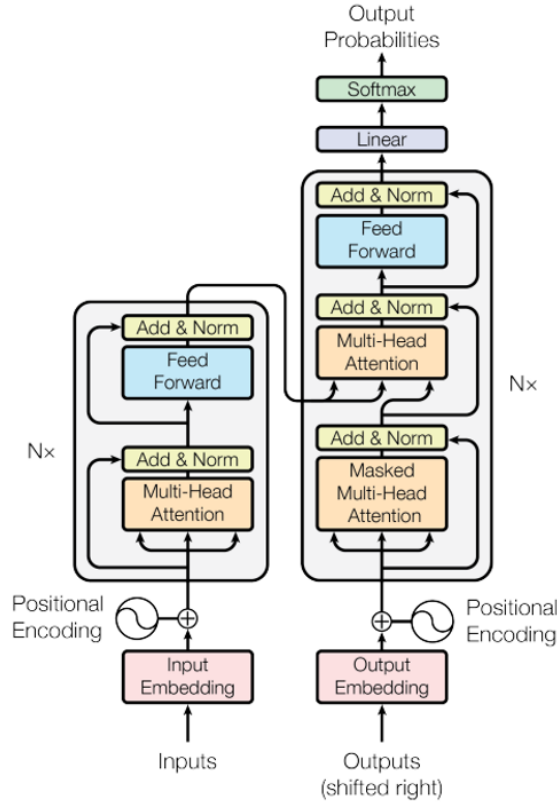


FIGURE 4.8: Encoder-Decoder Structure (Vaswani et al., 2017)

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right) \quad (4.5)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right) \quad (4.6)$$

where pos is the position,

i is the dimension.

This model demonstrated excellent performance on machine translation tasks for the first time and ushered in a new era of pure attention networks (Vaswani et al. 2017).

4.1.3.2 BERT(Bidirectional Encoder Representations from Transformers)

BERT is the first large-scale pre-trained language representation model proposed based on Transformer Encoder, with the following features: 1. Bidirectional MLM (Masked Language Model): randomly mask 15% of the input tokens, and the model needs to restore the original word from the left and right contexts at the same time to overcome the one-sided information of the unidirectional language model; 2. Next Sentence Prediction (NSP): train the model to determine whether two sentences are adjacent in the original text to enhance cross-sentence

understanding; 3. Parameter scale: BERT-Base contains 110M parameters (12 layers, 768 hidden dimensions, 12 heads of attention), BERT-Large contains 340M parameters (24 layers, 1024 hidden dimensions, 16 heads of attention); 4. Large-scale pre-training corpus: including BookCorpus (800M words) and English Wikipedia (2500M words), using the WordPiece tokenizer (Devlin et al. 2019). BERT sets new state-of-the-art results on multiple benchmark tasks such as GLUE and SQuAD, establishing the status of the pre-training + fine-tuning paradigm (Devlin et al. 2019).

4.1.3.3 RoBERTa(Robustly optimized BERT approach)

RoBERTa systematically optimizes BERT’s pre-training strategy: 1. Remove NSP tasks: According to experiments, it can be proved that NSP has limited improvement on downstream tasks and can be omitted; 2. Dynamic Masking: Randomly mask in each iteration to improve training diversity; 3. Larger pre-training corpus: Use CC-News, OpenWebText, Stories, etc., a total of 160GB of text; 4. Long training steps and large batches: Longer training cycles and larger batch sizes allow the model to fully learn; 5. Detailed hyperparameter tuning: including learning rate warm-up, weight decay, etc (Liu et al. 2019).

4.1.4 LLM

TABLE 4.1: Key Parameters of T5 Model

Parameter	Value
Encoder layers	12
Decoder layers	12
Hidden dimension	768
Multi-head attention heads	12
Feedforward network intermediate dimension	3072

T5-Base serves as the pre-trained backbone for this study, comprising 12 encoder layers and 12 decoder layers, with a total parameter count of approximately 220 million. Each layer has a hidden dimension of 768, and multi-head self-attention (MHSA) employs 12 attention heads, and the intermediate dimension of the feedforward network (FFN) is 3072. Each layer of the model is stacked in the order of

input $\rightarrow$ MHSA $\rightarrow$ residual connection + LayerNorm $\rightarrow$ FFN $\rightarrow$ residual connection + LayerNorm

(Raffel et al. 2020). This architecture enables parallel multi-head attention to capture multiple sub-space dependencies, feedforward networks to achieve nonlinear transformations, and residual connections and layer normalization to ensure gradient stability and information flow (Raffel et al. 2020). This unified Encoder–Decoder structure not only efficiently models long-range dependencies within sequences but also provides a solid modular foundation for instruction-based fine-tuning.

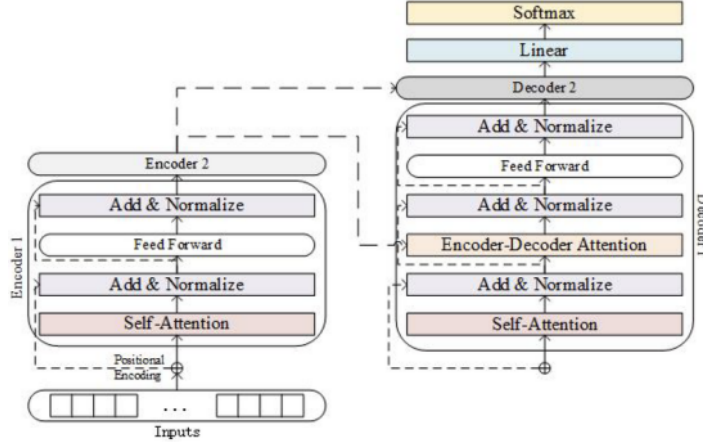


FIGURE 4.9: T5 Model Architecture.(Wang et al. 2023)

To address the issues of high memory consumption and low training efficiency caused by full fine-tuning in large models, this study introduces the LoRA (Low-Rank Adaptation) mechanism into the Encoder part of T5 to improve parameter efficiency and resource utilization during fine-tuning (Hu et al. 2021). Specifically, we inject LoRA into the core Query (Q) and Value (V) projection matrices of the Encoder Attention module. In the original T5 architecture, the Q, K, and V matrices in the Self-Attention module are typically mapped from the input vector via fully connected layers, resulting in a large number of weight matrix parameters (Hu et al. 2021). The core idea of LoRA is to introduce a residual path represented by a low-rank matrix while freezing the original pre-trained weights, thereby avoiding updates to the entire model parameters.

$$W = W_0 + AB, \quad A \in \mathbb{R}^{d \times r}, \quad B \in \mathbb{R}^{r \times k}, \quad r \ll d, k$$

Notation Explanation.

- W_0 : Original weights obtained from *pre-training* (frozen during fine-tuning);
- A, B : Low-rank trainable parameters;

- r : Rank control parameter, generally much smaller than d and k , thereby significantly reducing the number of trainable parameters.

To ensure numerical stability and training signal strength, LoRA also introduces a scaling factor α in its implementation, which is ultimately applied to the query path as a sample in the form of:

$$W_Q = W_Q^{(0)} + \frac{\alpha}{r} A_Q B_Q$$

- W_Q : Final query projection weight after LoRA injection;
- $W_Q^{(0)}$: Pre-trained (frozen) query projection matrix;
- A_Q, B_Q : Trainable low-rank matrices for query update;
- α : Scaling factor;
- r : Rank of the low-rank decomposition (typically $r \ll d, k$).

The above structure means that, while keeping inference latency and storage overhead almost unchanged, we can adapt to new tasks by training only a small portion of parameters (Hu et al. 2021), thereby significantly improving fine-tuning efficiency, especially for ABSA tasks with scarce labeled data and large cross-domain differences. In this experiment, the rank r of LoRA was set to 8, and the scaling factor α was set to 16. We observed a reduction in memory usage of approximately 35%, and a reduction in training time of approximately 40%, while maintaining good task performance on the SEMEVAL-2014 and SEMEVAL-2015 datasets.

Technical Solution Design and Comparative Analysis.

This project designed and implemented three different technical approaches for the ABSA (aspect level sentiment analysis) task, with the aim of identifying the optimal technical architecture to optimize model performance. The three approaches are as follows:

- (1) **BERT-based feature representation + T5 multi-modal classification head:** First, a pre-trained BERT model is used to perform tokenization and embedding of the input text as the basic semantic representation. Based on this, two independent classification branches are constructed using T5: one to predict the BIO sequence corresponding to aspect words, and the other to classify the sentiment polarity of the entire sentence. Finally, the outputs of the two branches are concatenated into a unified multitask classification head for joint training. A joint loss function is used for training, and LoRA is injected into the T5 encoder stage for fine-tuning.

- (2) **T5-based Generative Prompt Model:** This approach uses the complete T5 encoding-decoding architecture to transform the ABSA task into a natural language generation task. By constructing prompts such as “Extract aspect and sentiment: Sentence: ..., Aspect: ...”, the model directly generates a text with a BIO sequence and sentiment labels, such as {Aspect; Sentiment: }, and then parses the output to obtain the final labels. To improve training efficiency, LoRA is injected into the T5 decoder stage.
- (3) **T5 Encoder as a feature extractor + multitask classification head:** In this scheme, the T5 encoder is used as a context-aware text feature extractor, and the decoder part is removed. Based on the encoder output, a custom multitask classification head is connected to separately process aspect tagging (BIO) and sentiment polarity classification. A joint loss function is applied for end-to-end training.

Solution	Technical Approach	Advantages	Disadvantages	Macro F1
1	BERT + T5 + Multi-modal Head	Clear module separation	Inconsistent characterization; chaotic feature fusion; poor results	0.43
2	T5 + Prompt Generation	Flexible; supports new tasks and instructions	Generation instability; slow inference; error-prone parsing	0.62
3	T5 Encoder + Multi-task Classifier	Uniform features; compact structure; stable training	No decoder; slightly weaker in context modeling	0.70

TABLE 4.2: Comparison of Proposed Solutions on SEMEVAL-2014 (Macro F1)

Due to the small size of the SemEval-2014 dataset (approx. 6,000 entries), it was concatenated with SemEval-2015 to form a combined dataset for large model training.

4.2 Data Collection

Data selection process involves 3 main selection criteria.

- Data should relate to ASBA and suite the project requirements.
- Evaluation metrics and report visualizations between two selected datasets should clearly articulate the evaluation of NLP models.
- The data selected should not be biased and the test datasets should not influence the training of the models.

During initial phase, 3 datasets Twitter comments, Flip-cart shopping reviews and Semantic Evaluation datasets were analyzed and finally **Semantic Evaluation** datasets are shown for the project research.

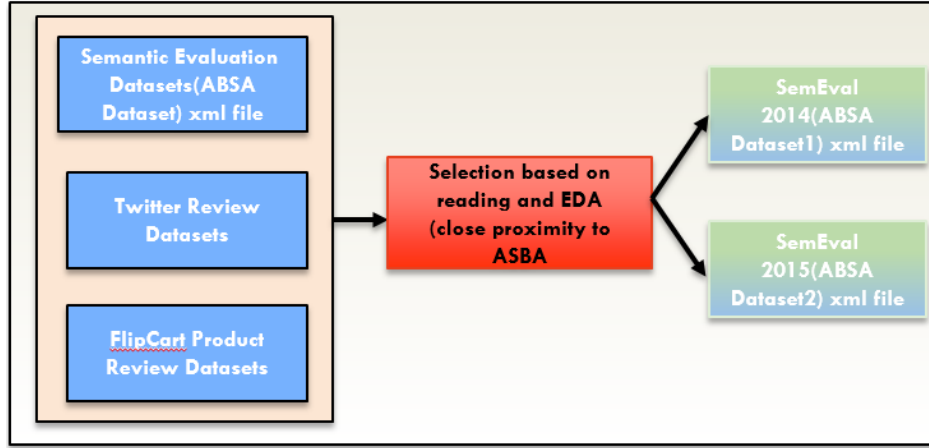


FIGURE 4.10: Data Selection Process

4.3 Data Analysis

Data pre-processing steps are illustrated in the diagram. Raw review xmls are converted into csv tabular form and review field are transformed further with string NLP manipulation techniques and tokenized before feeding into the model. Different tokenzier methods are used for different models to achieve runtime efficiency and results accuracy. For class labels, label encoder is used to convert text labels into (0 ,1, 2) integers.

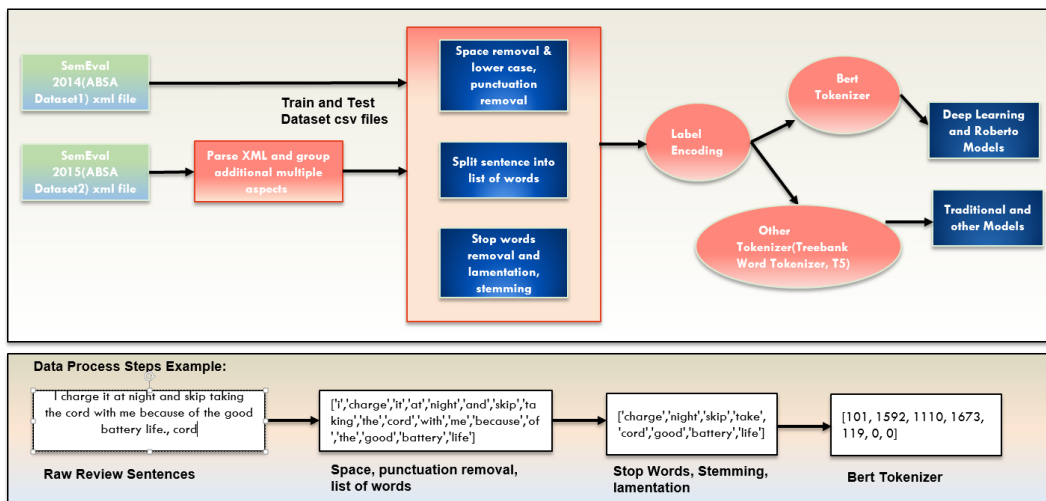


FIGURE 4.11: Data Pre-Processing Steps

SemEval 2014 and SemEval2015 released by SemEval for the subtask of aspect term extraction containing data from restaurant and laptop domains will be used for the project comparison study and demonstration of different NLP models and their evolution. Below table explains the features and label of each dataset. In the 2014 dataset, the review sentences are annotated using single aspect terms and their polarities. However in 2015 and sentence are described by aspect terms, aspect categories and their polarities.

Attribute Name	Attribute Description	Variable Type	Data Type	Sample Values
id	sentence id	Feature	Numeric	3121
Sentence	review comments for restaurants and laptops	Feature	Text	But the staff was so horrible to us.
Aspect Term	key aspect text	Feature	Text	staff
polarity	Category variable review polarity(positive, negative, neutral)	Label	Text	negative
from	main keyword aspect term start position of the review	Feature	Numeric	8
to	main keyword aspect term end position of the review	Feature	Numeric	13

TABLE 4.3: SemEval 2014 Dataset Features Description)

Attribute Name	Attribute Description	Variable Type	Data Type	Sample Values
reviewid	review unique id	Feature	Numeric	1004293
id	sentence id split for each review	Feature	Text	1004293:0
text	review comments for restaurants and laptops	Feature	Text	The food was lousy - too sweet or too salty and the portions tiny
target	key aspect text in multiple tags of sentences(multiple aspect)	Feature	Text	food portions
category	key aspect text in multiple tags of sentences	Feature	Text	FOODQUALITY
polarity	Category variable review polarity(positive, negative, neutral)	Label	Text	negative
from	main keyword aspect term start position of the review	Feature	Numeric	4
to	main keyword aspect term end position of the review	Feature	Numeric	8

TABLE 4.4: SemEval 2015 Dataset Features Description)

Exploratory data analysis for the datasets:

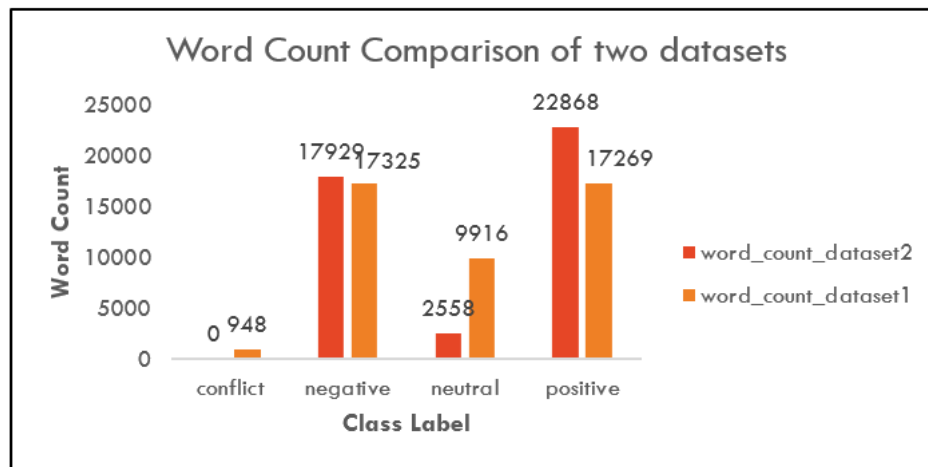


FIGURE 4.12: Data Selection Process

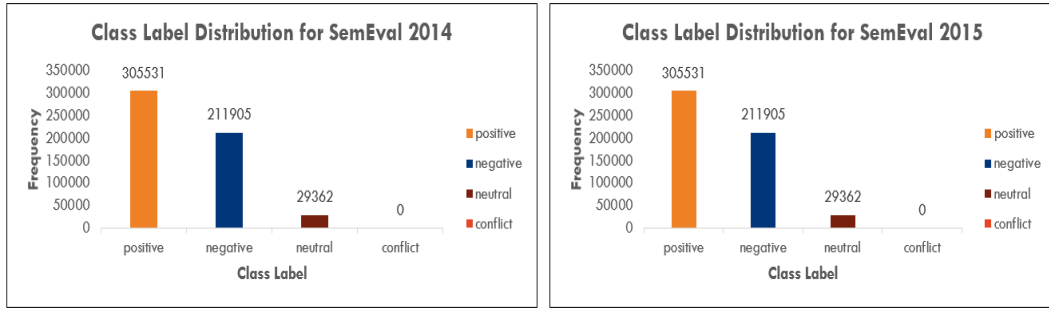


FIGURE 4.13: Dataset Label Distribution

4.4 Deployment

This project have no deployment process since the final product is a scientific report that covers all aspects in the project. Codes for each models are stored separately in the OneDrive folder shared to client with instruction to run individually.

4.5 Testing

There is no testing process needed for this project since as the final product is a scientific report. However, some testing are done during implementation of the models for optimization on performance. Still, no specific testing tools are involved. Instead of testing, **evaluation metrics** and **XAI implementation** should be mentioned here.

- Evaluation Metrics:** All models adopted their own selected evaluation metrics for model performance optimization. Despite the mixed use of evaluation metrics in every models for their own exploration and experimenting, in order to form a comparative analysis with unified standard, we have decided to use **Accuracy** and **Macro F1 score** as the evaluation metrics in comparative analysis. For Accuracy, it is a common and well-accepted baseline in comparisons, and its easy to understand what this metric represented. On the other hand, Macro F1 score gives a class-balanced view of performance. This method ensure the performance score are fairly reflect on all classes despite of the size. Through overall assessment and universal even-distribution performance condition, together they provide a balanced, standardized, and easily interpretable evaluation framework for comparing ABSA models.

- **XAI Implementation:** For traditional NLP, deep learning and LLMs, models of these approaches uses Local Interpretable Model-agnostic Explanations (**LIME**) to explain the local decision-making process of the model, revealing the feature basis and credibility behind each prediction. And for transformer-based models, we use Captum’s Integrated Gradients (IG) method: **Top-K Important Word Histogram** and **Cumulative Attribution Coverage Curve** for interpretability.

All models used the evaluation metrics and XAI implementation described above. This ensure a solid foundation for the comparative analysis using results from this models.

RESOURCES

5.1 Hardware & Software

The hardware and platform specifics to run each models are as below:

- **SVM & Decision Tree:** Jupyter Notebook, Scikit-learn version: 1.4.2, CPU: Apple M3 Pro, GPU: None
- **RNN, GRU, LSTM, BiLSTM:** Jupyter Notebook, Torch 2.5.1+cu121, CPU: AMD Ryzen 5 4600H, GPU: NVIDIA 1650
- **BERT & RoBERTa:** PyCharm2024.3.5 (Prof.), Win10 x64, CPU: AMD Ryzen 9 5900HX, GPU: NVIDIA RTX 3060
- **T5:** Jupyter Notebook, Win10x64, CPU: AMD Ryzen 95900 HX, GPU: NVIDIA RTX 3060

The hyperparameter used for each models are as below(traditional NLP models using all default settings of hyperparameter):

TABLE 5.1: Hyperparameters for Deep Learning Models

Hyperparameter	Value
BATCH_SIZE	32
LR	(Depends on model and dataset)
EPOCHS	15
WEIGHT_DECAY	1e-5
shuffle	true (train) / false (test)
embedding_dim	(Depends on model and dataset)
hidden_dim	(Depends on model and dataset)
SEED	42

TABLE 5.2: Hyperparameters for Transformer-based Models

Hyperparameter	Value
BATCH_SIZE	16
LR	2e-5
EPOCHS	10
WEIGHT_DECAY	1e-5
MAX_LEN	128
SEED	42

TABLE 5.3: Hyperparameters for LLMs

Hyperparameter	Model 1: BERT + T5 Head	Model 2: T5 Full + Prompt	Model 3: T5 Encoder + MTL Head
Backbone	bert-base-uncased + t5-base	t5-base	t5-base (only Encoder)
Maximum sequence length	128	128	128
Batch size	8	8	8
Learning rate	2e-4	2e-4	2e-4
Optimizer	AdamW	AdamW	AdamW
Dropout Rate	0.3	0.3	0.3
Epochs	15	15	15
Loss function	CrossEntropyLoss	CrossEntropyLoss	CrossEntropyLoss
LORA	r=8, lora_alpha=16, lora_dropout=0.1 Using in T5 Decoder	r=8, lora_alpha=16, lora_dropout=0.1 Using in T5 Decoder	r=8, lora_alpha=16, lora_dropout=0.1 Using in T5 Encoder

For software, the project uses Onedrive and Google Drive for documentation. GitHub for code version control. Jupyter Notebook for coding. Overleaf.com and Microsoft Word for writing documents and formatting them. Wechat and Slack for group communication. Outlook for email writing to contact client, tutor, and teaching team.

5.2 Materials

All materials used in this project are open and free to public.

5.3 Roles & Responsibilities

The roles and responsibilities for each team member are as below:

- **Albert Huang:** Project Manager, Researcher, Data Analyst, Recorder, Reporter. Oversee and manage the entire project, communicate with client, tutor, and teaching team on behalf of the group. Document everything related to the project, writing and

formatting reports. Research on ABSA, XAI to form literature review. Analyze the project result to conduct comparative analysis.

- **Aaron Zou:** Developer, Data Science, responsible for LLM. Make research and write LLM related parts in the report.
- **Austin Mi:** Developer, Data Science, responsible for Deep Learning. Make research and write deep learning related parts in the report.
- **Jiajian Pang:** Developer, Data Science, responsible for traditional NLP. Make research and write traditional NLP related parts in the report.
- **Weiguang Lu:** Developer, Data Science, responsible for Transformer-based and make research and write transformer-based related parts in the report.
- **Prem Kumar Rajendran:** Developer, Data Analyst , Pre-processing, CICD Admin Management. Responsible for constucting weekly presentation slides (group review part) and present. Also data collection, data analysis, and pre-processing. GitHub repository manager.

CHAPTER 6

MILESTONES SCHEDULE

Milestone	Tasks	Reporting	Date
Week-1	Group forming and early preparation project	None	28/03/2025
Week-2	Project requirement gathering, resource allocating, working environment building	Client Meeting for requirement gathering	07/03/2025
Week-3	Confirm change of project direction	Client Meeting for change of project direction	14/03/2025
Week-4	Design work plan and proposal draft	Online client communication for check form presentation	21/03/2025
Week-5 Proposal Report	Proposal Report Completed & Data-preprocessing and model design	Client Meeting for progress update and proposal feedback	28/03/2025
Week-6	Development Week 1	None	06/04/2025
Week-7	Development Week 2	None	13/04/2025
Week-8	Development Week 3 Testing, Evaluation, Analyzing	Client Meeting for development stage and 2nd checking form	17/04/2025
Week-9 Implementation of all models	Progress Report submitted & Optimizing models	Email communication with client to update status	03/05/2025
Week-10	Optimizing models & collecting results & writing draft structure of scientific report	Email contacting client for status update and feedback	07/05/2025
Week-11	Documentation Finalization. Working on solving challenges in scientific reports	Communicate with client by emails and gained feedback	15/05/2025
Week-12	Complete draft of scientific report	updating client by emails	21/05/2025
Week-13 Scientific Report & Project Demo to Client	Delivered project to client with demo	Client Meeting for demo and final checking form	29/05/2025

RESULTS

First, from the results of literature review of ABSA evolution, we created a timeline graph based on our research.

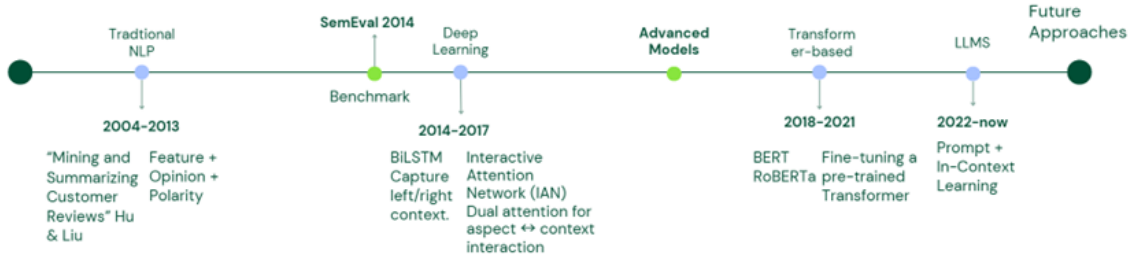


FIGURE 7.1: ABSA Evolution in Approaches Timeline Graph

In the perspective of evolution overview, ABSA evolved from feature engineering to representation learning, then into pre-training methods, and adapt prompting as the latest approach. Along with the evolution of NLP approaches and models, the performance using these techniques in ABSA keeps growing and the granularity is also improving. Lesser manual effort are needed as the approaches evolved and richer semantics can be captured. For the future, directions like LLM distillation for mobility, multi-modal ABSA and deeper integration with XAI to increase interoperability can gain high value in the applications.

On the practical experiment side, by adopting the methodologies as described in the above section, we implemented nine different models for conducting ABSA on the two selected dataset, SemEval 2014 and SemEval 2015. The nine models are: SVM, Decision Tree, RNN, GRU, LSTM, BiLSTM, BERT, RoBERTa and T5, which are selected on various reasons described in section 4, representing iconic ABSA approaches in the field of Traditional NLP, Deep Learning, Transformer, and LLM. With XAI implementation on each model, we are enabled to identify vital features, extra token sentiment polarity, validate model decisions...

and increase our understanding of the model working mechanism. We selected few iconic XAI visualizations from each section of approaches for demonstration and discussion here.

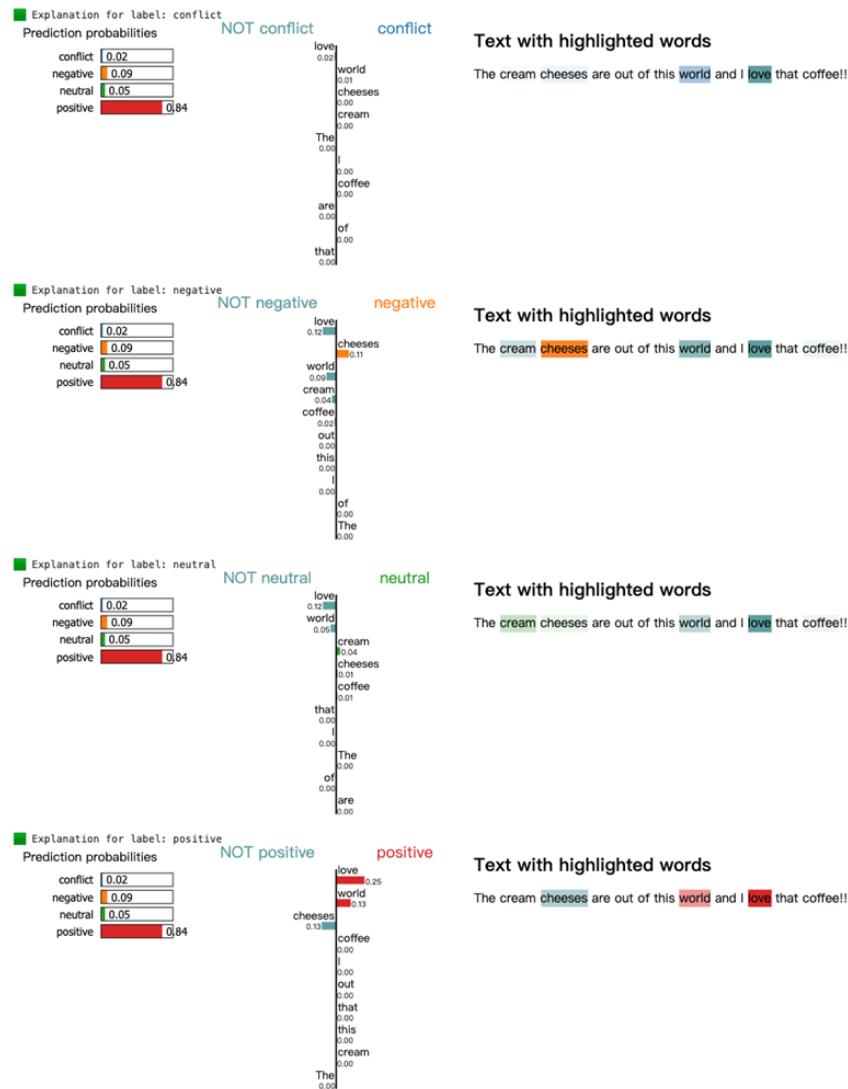


FIGURE 7.2: XAI Result from SVM Model (partial)

In the SVM and Decision Tree model, XAI enables us to gain a transparent view into the decision making of sentiment polarity. In the example sentence, some words are highlighted with different colours to infer their influence of the overall sentiment classification. For example, in the first graph, words like good, fresh contribute significantly towards NOT negative, affecting the prediction probabilities of the entire sentence. It seems that with XAI implementation, traditional NLP models like SVM and Decision Tree can be interpreted clearly for its decisions.

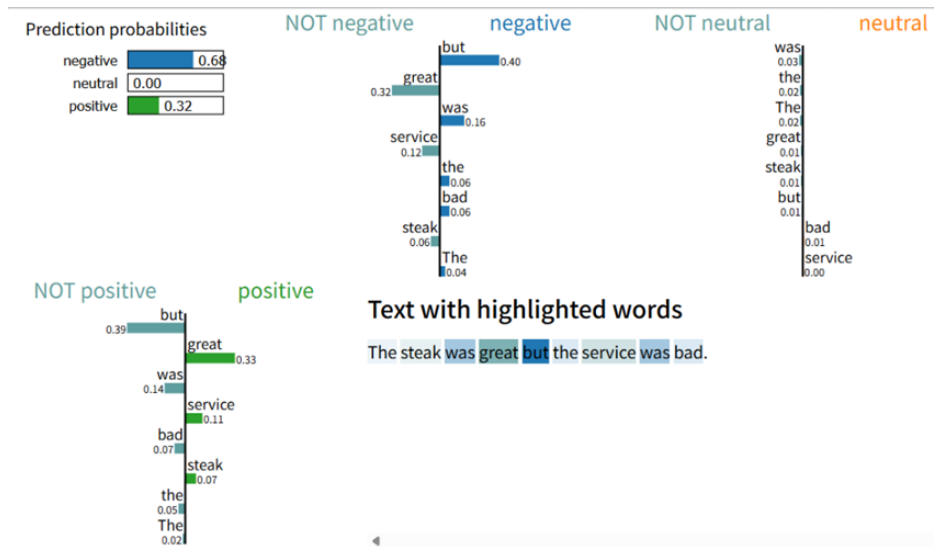


FIGURE 7.3: XAI Result from BiLSTM (partial)

In the BiLSTM model, XAI enabled us to gain insights into the model like the general local explanation of positive class here, which shows attribution weights of tokens in positive or negative, reflecting the contrastive conjunction ability of the model. In the next two example sentences, similar to XAI result in traditional NLP models, words in a sentence with highlights infer their contribution to the three classes of sentiment. Those visualizations illuminate the BiLSTM aspect-level prediction explanation of integrating both lexical sentiment cues and syntactic markers. XAI here gave us transparency and room for error-analysis to some degree.

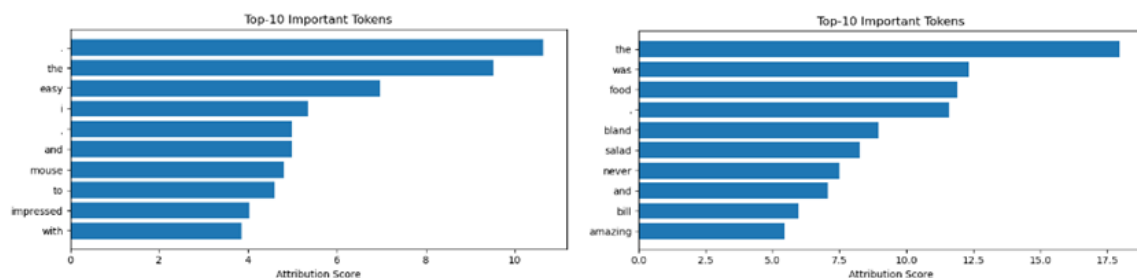


FIGURE 7.4: XAI results for BERT part 1 SemEval2014

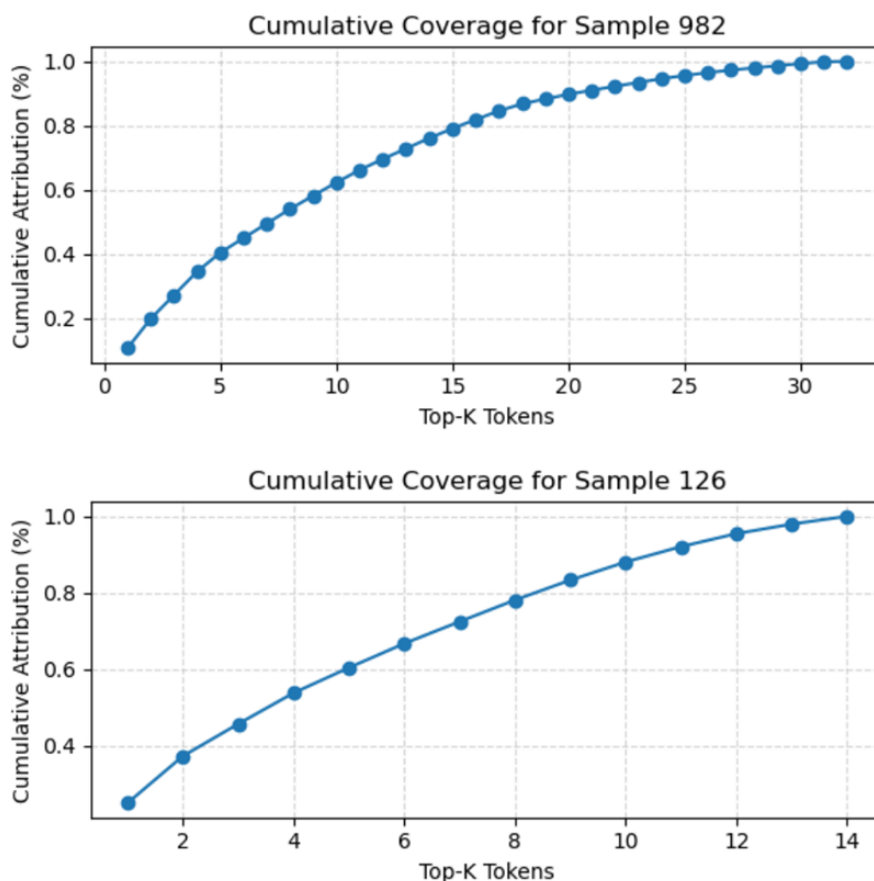


FIGURE 7.5: XAI results for BERT part 2 SemEval2014

For the BERT model, the XAI results above show the top 10 important tokens for a sample and its cumulative coverage of attribution in top-K tokens. For example, in sample 126, top 10 tokens like “the” “was” “food” “.” ... can already explain approximately 0.83 of the model’s decision rationale. And we can obtain the nearly complete coverage point of top-K tokens for these visualizations. XAI here provides model decision transparency in the terms of explaining influential tokens effects.

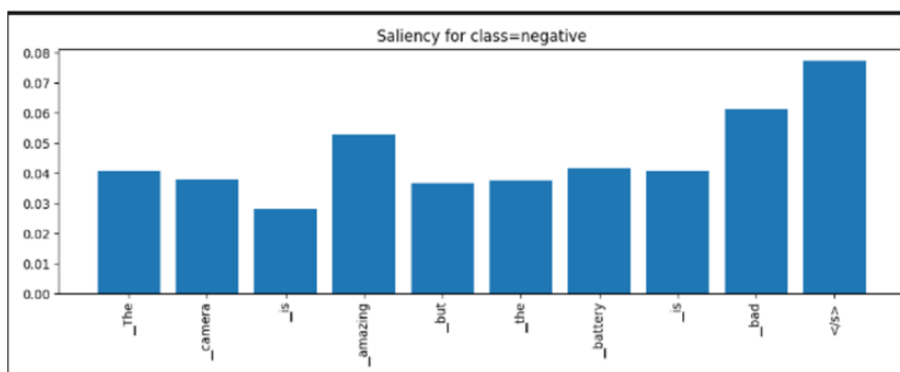


FIGURE 7.6: XAI Result from T5 Part 1

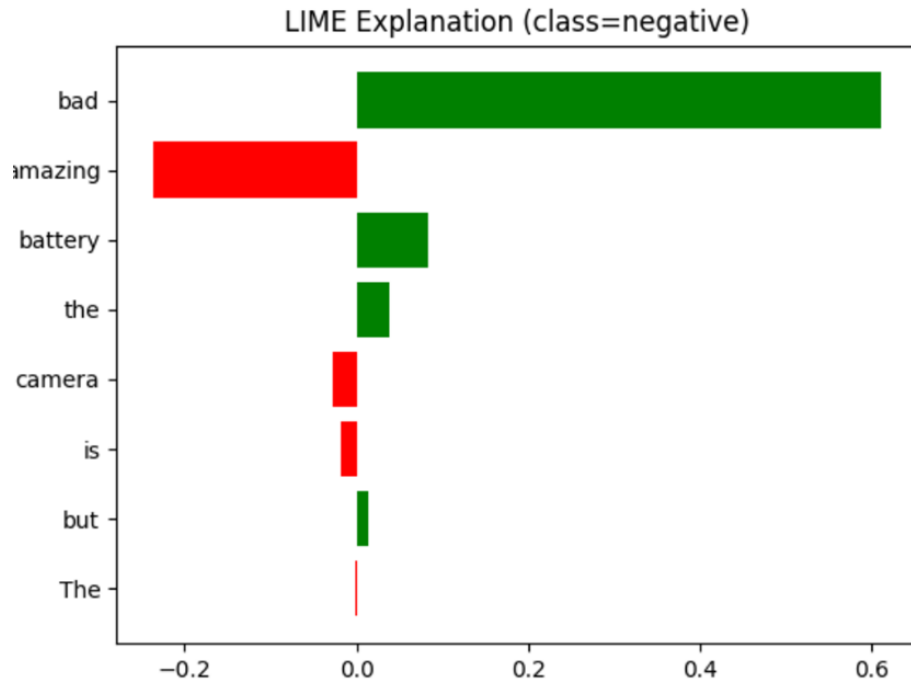


FIGURE 7.7: XAI Result from T5 Part 2

In T5 models, XAI results presented here use a classic example of conflictive emotion sentence “The camera is amazing but the battery is bad”. In the first graph, XAI enables us to clearly view the contribution of each token in the sentence towards the negative-class probability. And in the second visualization, gradient magnitudes over input embeddings are shown under the class of negative. Sequence end token and “bad” here register the highest saliency, which indicates possible weighting preference of final token and sentiment adjectives for this model’s prediction. XAI results here give information of model’s decision making, however, less explainable when compared to other models.

Apart from the result of XAI implementation on each model, by following the two selected evaluation metrics Accuracy (ACC) and Macro F1 Score, we collected results from all models and organized them into the comparative performance table below:

TABLE 7.1: Models Performance Comparison

Model	Dataset	Domain	ACC	Macro F1 Score
Traditional (SVM)	SemEval 2014	Laptop	0.72	0.51
		Restaurants	0.73	0.53
	SemEval 2015	Laptop	0.95	0.90
		Restaurants	0.94	0.84
Traditional (Decision Tree)	SemEval 2014	Laptop	0.66	0.46
		Restaurants	0.67	0.52
	SemEval 2015	Laptop	0.95	0.90
		Restaurants	0.95	0.88
Deep learning (RNN)	SemEval 2014	Laptop	0.4320	0.3085
		Restaurants	0.6103	0.3426
	SemEval 2015	Laptop	0.5510	0.5372
		Restaurants	0.6300	0.6037
Deep learning (GRU)	SemEval 2014	Laptop	0.6674	0.6389
		Restaurants	0.6963	0.5942
	SemEval 2015	Laptop	0.9560	0.9558
		Restaurants	0.9575	0.9574
Deep learning (LSTM)	SemEval 2014	Laptop	0.6199	0.5132
		Restaurants	0.6546	0.5267
	SemEval 2015	Laptop	0.9540	0.9535
		Restaurants	0.9560	0.9558
Deep learning (BiLSTM)	SemEval 2014	Laptop	0.6911	0.6693
		Restaurants	0.6893	0.6057
	SemEval 2015	Laptop	0.9580	0.9578
		Restaurants	0.9595	0.9594
BERT	SemEval 2014	Laptop	0.9143	0.9907
		Restaurant	0.9260	0.9925
	SemEval 2015	Laptop	0.9363	0.9616
		Restaurant	0.9550	0.9931
RoBERTa	SemEval 2014	Laptop	0.9314	0.9900
		Restaurant	0.9540	0.9945
	SemEval 2015	Laptop	0.9463	0.9619
		Restaurant	0.9625	0.9934
T5	SemEval 2014	/	0.83	0.70
	SemEval 2015	/	0.94	0.89

DISCUSSION

By implementing **XAI** in all models, we achieved a certain degree of transparency in models' sentiment polarity prediction decision-making process, though in various degrees. It can be seen that each model has its unique way of predicting. XAI quantifies the contribution of tokens, and shows the processing of models towards various linguistic situations. Overall, XAI could be of great help to model explanation and ability for debugging.

For **models performance**, on the level of approaches, models that adopted transformers gained the best performance on all datasets, reaching high scores on both ACC and Macro F1 (average 0.95+ on all entities involved). This indicates the great aspect-level sentiment classification capability of transformer models, which might be the effect of their self-attention mechanisms. The modern architecture of these models allows them to capture dependencies with aspect terms and sentiment modifiers far better than traditional recurrent models. However, T5 as another modern advantage model received a worse outcome than transformer models on both datasets, and on the semeval 2015 dataset, gained a overall worse score on both ACC and Macro F1 than three deep learning models (GRU, LSTM, BiLSTM). This outcome is suspected to be caused by the special procedure adopted for T5 (referred to section3) as both Laptop and Restaurant are combined as one for training due to the resource limitation on this project. Moreover, the high resource consumption for running such a large model leads to necessary change of structure to fit this study. All these factors might cause the T5 model not reaching its full potential. Nonetheless, T5 still shows good performance overall when compared to other traditional or deep learning approaches, indicating the superiority of advanced models in ABSA. Traditional models display a similar level of performance to deep learning models. With some specific models achieving high scores and some low scores, traditional models and deep learning models also displayed different performance when encountering different datasets.

On the level of models, the best model of each approaches judging from their ACC and Macro F1 score for both datasets are: SVM (traditional NLP), BiLSTM(deep learning), RoBERTa (transformer) and T5 (sole model for LLM). These models should be prioritized to be considered for an ABSA task within its own type of approach. RNN is the worst model of all, with low scores on both datasets. For this situation, there are few possible reasons: poor long-range dependency modeling, fixed recurrence failed to capture nuanced aspect-sentiment interactions. And with the far better performance of other deep learning models, it also shows the importance of having gating or attention mechanisms for ABSA. What also needs to be mentioned is that within these models, the performance on dataset Semeval 2015 is better overall than dataset Semeval 2014. This could be caused by differences in quality and annotation consistency, vocabulary alignment, class balance... between the two datasets. Those factors of dataset properties affect the performance of a model. However, despite the difference in dataset, models show a similar rank of performance across these dataset, indicating the comparative evaluation of model performance is not affected by datasets. Thus, the results of comparison are general among these models. This also indicates the importance of data quality for ABSA tasks.

In summary, despite the experimenting attempts on LLMs, transformer-based architecture shows significant lead in ABSA in this study, which could reflect on larger, different datasets with the same trend of performance here. Advanced models overall show superiority over traditional approaches. Traditional models like SVM remain competitive along with deep learning models with gating or attention mechanisms. They can be a good choice under limited resources or other special circumstances. Great model architecture and high quality of dataset is the key to produce a reliable ABSA performance.

Besides comparison on performance, we also conducted in-depth exploration on each model, using XAI as a tool of interpretation. We summarized the advantages and disadvantages of each model under different circumstances, along with comparing the complexity, explainability and resource consumption. The outcome of our experiment is shown in the table below.

Model	Complexity	Explainability	Resource Consumption	Con-
Traditional (SVM)	Low. Simple model structure with few parameters.	High. Clear linear decision boundaries.	Very low.	
Deep learning (RNN, GRU, LSTM, BiLSTM)	Overall medium-high. RNN simplest; GRU adds gates; LSTM adds memory cell; BiLSTM bidirectional, most complex.	Overall medium-high. RNN easiest to interpret; GRU/LSTM harder; BiLSTM lowest explainability.	Medium. Resource use: BiLSTM $\gg$ LSTM $>$ GRU $>$ RNN	
BERT	High. 12-layer Transformer (110 M parameters).	Low-medium. Self-attention helps but deep stacking hinders visibility; needs post-hoc XAI.	High. Requires several GB GPU memory and has noticeable latency.	
RoBERTa	High. Like BERT-base but 125 M (base) / 355 M (large) params; more pre-training.	Low. Even deeper than BERT, interpretability further reduced; relies on post-hoc XAI.	Very high. Pre-training fine-tuning inference all more resource-hungry.	
T5	Very high. Full encoder-decoder, T5-base $\approx$ 220 M parameters	Low-medium. Complex autoregressive encoder-decoder makes XAI difficult.	Very high.	

TABLE 8.1: Overall Models Comparison part 1

Model	Advantage / Disadvantage
Traditional (SVM)	Advantage: easy to implement; fast training; performs well on small datasets. Disadvantage: limited generalization; weak on complex semantic relationships.
Deep learning (RNN, GRU, LSTM, BiLSTM)	Advantages: RNN simple and quick; GRU matches LSTM performance with higher efficiency; LSTM good long-term memory and generalization; BiLSTM best overall performance. Disadvantages: RNN struggles with long dependencies; GRU's gating less fine-grained than LSTM; LSTM slow due to many parameters; BiLSTM very resource-intensive, low inference efficiency.
BERT	Advantage: excellent context encoding; strong performance across NLP tasks; easy to fine-tune. Disadvantage: large model size; slow training/inference; high GPU/CPU usage.
RoBERTa	Advantage: more robust pre-training strategy; outperforms BERT on many benchmarks. Disadvantage: even greater training/deployment overhead; slower inference; extremely high resource demands.
T5	Advantage: unified, flexible encoder-decoder architecture; high development efficiency; strong generalization. Disadvantage: very large; slow training/inference; extra engineering effort for ABSA; output consistency issues.

TABLE 8.2: Overall Models Comparison part 2

LIMITATIONS AND FUTURE WORKS

In conclusion, this project has provide entrance-level-understandable readability for new researchers into the field. It should provide them with comprehensive yet concise view on each types of approaches throughout the evolution of ABSA. In-depth exploration and comparison on each signature models are conducted and results discussed.

However, with the resource constrains on this project, only a limited number of models can be implemented and a limited scale of datasets can be selected. The performance of the models can be affected by the datasets, XAI implementation and hardware (advanced, complex models). Ideally the result of the project should be able to reflect on more complex, closer-to real world scenarios. But with project's limitations, the result of this project should be considered accordingly outside the environment in this project.

For future works, more advanced models like Llama3 can be implemented and larger datasets can be used to widen the range of this project and bring in more variables for analyzing. Besides approaches experimented in this study, other new emerged approaches like hybrid models which combined multiple techniques or models can attain better performance than single models on some specific ABSA tasks. Multi-model solution that combines not only text-input but also images, audios and more is also an important direction at the moment and in future studies. The ABSA field has a promising future with dynamic research direction for any who are interested.

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